- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

8 A bus is travelling along a straight road at $5.4\,\mathrm{m\,s^{-1}}$. At $t = 0$, as the bus passes a boy standing on the pavement, the boy starts running in the same direction as the bus, accelerating at $1.2\,\mathrm{m\,s^{-2}}$ from rest for 5 s. He then runs at constant speed until he catches up with the bus.

(a) The diagram in the Printed Answer Booklet shows the velocity-time graph for the bus.

Draw the velocity-time graph for the boy on this diagram. **[3]**

(b) Determine the time at which the boy is running at the same speed as the bus. **[2]**

(c) Find the maximum distance between the bus and the boy. **[3]**

(d) Find the distance the boy has run when he catches up with the bus. **[3]**

